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Effective transfer learning of defect detection for photovoltaic module cells in electroluminescence images☆

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ABSTRACT

Due to the damage during production, transportation and installation, some defects inevitably occur in the solar cells, which will reduce the power generation efficiency. Benefiting from the development of deep learning, the performance of solar cell defect detection has been improved by a considerable margin. However, a problem persists that a model trained on one batch of data may not work on another batch following different production schemes. Moreover, compared with monocrystalline cells, more impurities on the surface of polycrystalline cells make it more difficult to inspect defects and obtain labels. To address this problem, we formulate it as a monocrystalline to polycrystalline task based on an unsupervised domain adaptation task in transfer learning. A standard network with CNN architecture is adopted as our classifier, which can tell whether the solar cell is cracked. Based on that, we embed adversarial learning in the network, where the domain discriminator is added to distinguish data domains, so that the feature extractor can learn domain-invariant features. Considering the scenario of defect detection in Electroluminescence (EL) images, we further propose attention-based transfer learning and class-aware domain discriminator to enhance the effect of knowledge transfer. When detecting a batch of new polycrystalline products, our method can improve F1-score by 0.2631, and the recall and precision can reach 84.70% and 90.15%, respectively.

1. Introduction

With the popularization of the sustainable development concept, solar energy has become the focus of energy structure optimization due to its unique advantages such as being inexhaustible, pollution-free and cheap. The application scope and scale of photovoltaic (PV) power generation are also expanding rapidly. The core working structure of PV power generation is the PV module composed of crystalline silicon solar cells connected in series or parallel. The working status of PV modules will directly affect the power generation efficiency.

The rapid development of the photovoltaic industry has promoted the increase in material costs for solar cell production, and solar cells have become thinner to reduce silicon consumption. The crystal structure of the thin silicon wafer itself makes it easy to crack, and the thinning of the solar cells reduces the ability of components to prevent mechanical damage, resulting in cracks of various sizes and shapes in the cells (Deitsch et al., 2019). If the cracks are not found in time, the cells may be broken when being subjected to pressure in the subsequent manufacturing process, resulting in increased production costs. Moreover, the hazard of micro-cracks may increase over time and ultimately affect the efficiency and reliability of the PV modules (Köntges et al., 2011; Kajari-Schršder et al., 2012).

To accomplish the quality detection of solar cells more intuitively, some methods for visualizing the internal structure of solar cells have emerged, such as Infrared imaging (King et al., 2000), Photoluminescence imaging (Redinger et al., 2016), and Electroluminescence (EL) (Fuyuki and Kitiyanan, 2009; Lockridge et al., 2016). Currently, EL imaging is widely used due to its advantages of non-contact, robustness, fast speed, and high-resolution imaging. It applies positive direct current voltage to the cells and uses the silicon charge-coupled device sensor, which helps image the near-infrared light emitted by the cells more quickly. Besides, EL imaging can also reveal the internal structures of the cells for defect detection (Ory et al., 2021).

However, it is still laborious and subjective to detect the crack defects of cells only by observing EL images manually. The inefficiency of

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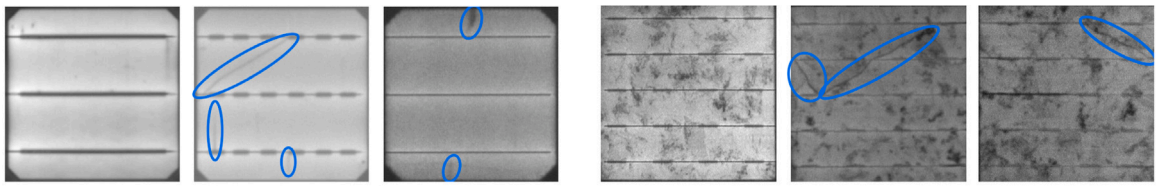


Fig. 1. Examples of EL images of monocrystalline and polycrystalline cells. Among the three EL images listed in each category, the first one is the cell without cracks, and the next two are the cell with cracks, in which the cracks are marked with blue circles.

defect detection makes it difficult to be applied to large-scale products. Moreover, there exist some micro-cracks that are tough to identify with the naked eye.

To solve the barriers of manual detection, many works introduce automatic defect detection methods, which perform well for EL images in different scenarios. At an early stage, many defect detection approaches based on traditional image processing were developed. Tsai et al. (2012) used the Fourier image reconstruction method to identify line- or bar-shaped defective regions. Anwar and Abdullah (2014) utilized anisotropic diffusion filtering and shape analysis for defect detection of solar cells. Tseng et al. (2015) leveraged binary clustering of candidate region features to detect finger interruption defects. Recently, the success of deep learning has been leading to a gradual replacement of traditional pattern recognition pipelines for visual perception tasks. Deitsch et al. (2019) introduced hardware-demanding convolutional neural networks (CNN) for the automatic classification of defective PV module cells in EL images. Tang et al. (2020) proposed a data augmentation method based on geometric morphometric characteristic to expand the existing EL images to better train the CNN-based defects classification model. Shou et al. (2020) proposed defect detection for EL images of solar cells based on the generative adversarial network to alleviate the problem of insufficient abnormal samples. Demirci et al. (2021) used the machine learning (ML) method to classify the image features extracted by deep neural network, and designed a lightweight CNN structure, which also obtained competitive results.

Recently, CNN-based methods have come to the fore in the solar cell defect detection community. In general, the CNN model is always trained on a fixed batch of data. Once the training is over, it can only be certain that the model works only on similarly distributed data. But in practical application, due to the differences between production technology schemes, the data distributions of cells can be variant. For example, the monocrystalline and polycrystalline cells are crystalline silicon cells under different cell technologies, and their appearance is significantly different. Cells from another data distribution are not suitable for existing models, *i.e.*, **model trained on one domain cannot be directly transferred to another domain.** In this case, another model has to be trained to adapt to this kind of domain shift. For the training of this new model, labeling new data is also needed, which is time-consuming and labor-intensive.

Transfer learning offers a possible solution for this cross-domain learning problem. The scenario can be formulated as a representative problem in transfer learning: unsupervised domain adaptation (UDA). With this problem setting, the training data is considered to be sampled from two different domains: labeled data from the source domain and unlabeled data from the target domain. Our aim is to train a model which works in both domains. The key is how to transfer the classification ability obtained on labeled data from the source domain to the target domain.

In this work, we are devoted to detecting defects in crystalline silicon cells. Crystalline silicon cells can be divided into monocrystalline cells and polycrystalline cells, as shown on the left and right of Fig. 1, respectively. The model transfer can be on the same type of products or between different types. As shown in Table 1, the detection ability of the model based on ResNet-50 works in the same domain, but the cross-domain effect is obviously worse. Besides, because the surfaces of the polycrystalline cells are not as clean as that of the monocrystalline cells, it is hard to obtain accurate polycrystalline data labels. Experiments and hard labeling issue show that the transfer from monocrystalline products to polycrystalline ones is very useful in practical applications, and it is still a problem to be settled.

Table 1
The F1-score results of the ResNet-50 model under different source domain to target domain scenarios. The detection ability of the model trained on monocrystalline cells does not perform well when applied to polycrystalline ones, and there is still room for research.

Source domain to target domain	F1-score
Monocrystalline to monocrystalline	0.7143
Polycrystalline to polycrystalline	0.8333
Monocrystalline to Polycrystalline	0.6103

Currently, there are many effective methods for UDA. Most of them are attempting to align the distributions of source and target domains by learning domain-invariant representation. They are focusing on directly minimizing the discrepancy between feature distributions of two domains or adversarially learning to enforce the feature representations to be indistinguishable by a domain discriminator. But there are two main problems if we directly apply existing approaches to EL image defect detection. First, for the academic dataset, category information is richer, so the feature is more distinguishable, reducing the difficulty of minimizing feature distribution discrepancy. However, the defect detection problem is a binary classification problem. A lot of redundant information is included in the classification features. There is a chance that two domains can be better aligned in a certain subspace of the classification feature. In this case, this redundant information is harmful to the domain alignment of the classification features. Another problem is also caused by the small number of categories. Under this circumstance, directly aligning the features of positive and negative samples indiscriminately makes it more possible to fail to learn knowledge useful for classification. The whole transfer process cannot be class-agnostic.

To address the above problems, we take two aspects into consideration: **what to transfer** and **how to transfer**. For the first aspect, we need a more clear indicator that can help us learn domain-invariant features. It occurs to us that the attention mechanism may contain the information we need. Following this idea, we embed the channel attention in our network and propose attention-based transfer learning. For the second aspect, we make the transfer process class-aware. Since the data from the target domain is unlabeled, classifier predictions are used during training as a soft form of pseudo-label, enabling the class-aware transfer on both domains.

Detailed experiments show that the simultaneous migration of the output features of each attention mechanism structure and the final features of the network can most effectively improve the adaptability of the model and improve the F1-score of the model by 0.2896. After further introducing the class-aware domain classifiers, the F1-score of the model can reach 0.8734. The contributions of this paper are summarized as follows:

1. We formulate defect detection for PV module cells as an unsupervised domain adaptation problem for cross-domain classifier training based on EL images.
2. We propose attention-based transfer learning and class-aware domain discriminator to enhance the feature learning for PV module cells in EL images.
3. Detailed experiments show that our proposed method can promote the performance of defect detection from monocrystalline domain to polycrystalline domain.

2. Related work

2.1. Defect detection of solar cells

For the defect detection problem, there are mainly two methods of imaging: Photoluminescence (PL) (Trupke et al., 2006; Johnston et al., 2013) and EL (Fuyuki and Kitiyanan, 2009). Since some defects in solar cells are only displayed in EL imaging of PV modules, the current mainstream defect detection methods for solar cells adopt EL (Mayr et al., 2019; Balzategui et al., 2019; Spataru et al., 2016).

In Tsai et al. (2012), a Fourier image reconstruction method was proposed to detect line- or bar-shaped defects in EL images of solar cells. Due to its shape assumption, it is difficult to detect defects with different shapes. Then, Tsai et al. (2013) proposed to use images based on Independent Component Analysis (ICA) to identify defects, where each solar cell sub image is reconstructed as a linear combination of the learned basis images. Anwar and Abdullah (2014) proposed the use of anisotropic diffusion filtering and shape analysis to detect defects in solar cells, which also realized defect localization. Tseng et al. (2015) used binary clustering of candidate region features to detect defects for finger interruption. Deitsch et al. (2019) introduced hardware-efficient SVM and hardware-demanding CNN for the automatic classification of defective PV module cells in EL images. It shows the great advantage of CNN architecture for classification tasks, including defect detection. In Balzategui et al. (2019), CNN was also adopted and helped suggest those cells that are defective for re-inspection. In the meanwhile, a rough segmentation map can also be provided. Mayr et al. (2019) were devoted to the segmentation of cracks on solar cells. They employed L_p norm to aggregate different layers of activation maps into a single value for classification and set an adaptive threshold to transform feature maps into segmentation maps. Tang et al. (2020) presented a data augmentation method based on the geometric morphometric characteristic and then designed a defects classification model of monocrystalline silicon cells EL images based on CNN by extracting these features, and expanding the existing EL images to train a better classifier. To solve the problem of insufficient abnormal samples, Shou et al. (2020) detected defects in the EL images based on the generative adversarial network, and use normal images in the training process. Pratt et al. (2021) proposed a model based on semantic segmentation to generate pixel-wise classifications for different features and defects, the model performed better on larger intrinsic features such as cell spacing and banded interconnections compared to the smaller extrinsic defects such as cracks and gridline defects. Demirci et al. (2021) proposed to use of different ML methods to classify the image features derived by the activations obtained from the middle activation layers of the networks concatenated in different combinations. Meanwhile, they proposed a lightweight CNN structure, which can also show competitive defect detection ability.

For our method, we focus on the problem of domain adaptation in defect detection of solar cells coming from multiple production lines based on diverse technical schemes. The transfer learning method is adopted to enhance the transfer effect.

2.2. Unsupervised domain adaptation

Unsupervised domain adaptation aims to transfer the knowledge from a labeled source domain to an unlabeled target domain. Abundant UDA works focus on the image classification task. The mainstream approaches tend to address UDA by domain alignment and discriminative domain-invariant feature learning. They mitigate the distribution divergence between the source and target domain and pull them into the same distribution at different scale levels.

Maximum Mean Discrepancy (MMD) (Borgwardt et al., 2006) is a widely used distribution discrepancy metric to measure domain similarity. Deep domain confusion (Tzeng et al., 2014) utilized MMD after an extra adapt fully connected layer to learn both discriminative and transferable domain features. Long et al. (2015) proposed a deep adaptation network (DAN) to minimize MMD over domain-specific layers. In Long et al. (2017), the joint MMD over multiple domain-specific layers is introduced to a joint adaption network to align the joint distribution of different domains. Yan et al. (2017) argued that ignorance of class weight bias can deteriorate the domain adaptation performance. So they proposed weighted MMD to re-weight the source domain, the weight of which is calculated through the classification EM algorithm. Some methods (Sun et al., 2016; Sun and Saenko, 2016; Peng et al., 2019) also introduced metrics based on some higher-order statistics. Sun et al. (2016) proposed to align the feature distribution of source and target domains with covariance matrix. In the actual implementation, the source feature is first whitened and re-colored with target covariance. The deep CORAL (Sun and Saenko, 2016) applied the covariance alignment to the deep network, where the whitening and re-coloring process can be got rid of. In Peng et al. (2019), the moment was introduced as another higher-order discrepancy metric to enable multi-source domain adaptation.

Another popular type of methods are devoted to learning domain-invariant representation using adversarial training. DANN (Ganin and Lempitsky, 2015) was the first work that leveraged adversarial learning in UDA. It proposed to train a domain discriminator that can distinguish the source domain from the target domain, meanwhile, a feature extractor is trained to fool the discriminator so that the feature distributions of the two domains are aligned. CDAN (Long et al., 2018) designed two novel conditioning strategies: multilinear conditioning and entropy conditioning to improve discriminability and guarantee transferability in the meantime. MCD (Saito et al., 2018) took task-specific decision boundaries into consideration. It first detects target samples that are far from the support of the source via maximizing the discrepancy between two classifiers' outputs. Then the discrepancy can be minimized with target features near the support. SymNets (Zhang et al., 2019) designed symmetric object classifiers for source and target domains, which are also used as domain discriminator at the same time and restricted by confusion losses. STAR (Lu et al., 2020) modeled the classifier as a Gaussian distribution with its variance representing the inter-classifier discrepancy, so that far more classifiers than before can be sampled. GVL (Cui et al., 2020b) balanced adversarial training through constructing bridge layers on both the generator and discriminator. MetaAlign (Wei et al., 2021) embedded meta-learning in the domain alignment objective and classification objective and optimized them in a coordinated way to alleviate the optimization inconsistency problem between domain alignment and classification tasks.

In our method, we also adopt the idea of adversarial learning and further design reasonable transfer mechanisms to better adapt to the scenario of defect detection in EL images.

3. Methodology

In general, the extraordinary feature representation ability of CNN provides a simple but effective way to solve the micro-crack detection for PV module cells. However, the diversity of technical schemes leads to different appearances and possible defects of various products, which

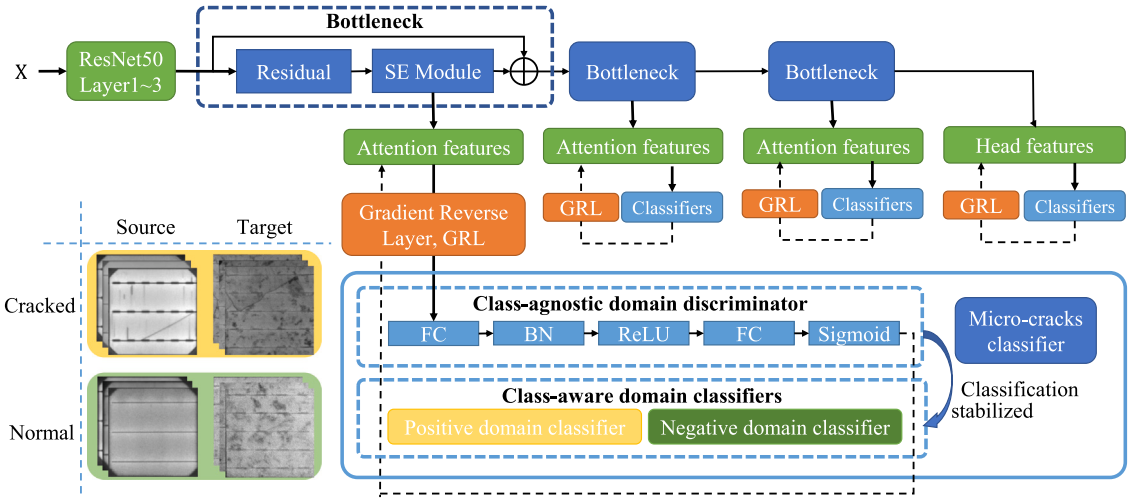


Fig. 2. The overall flow frame diagram of our method. The output features of the three attention mechanism modules and the final features of the whole network are aligned by using GRL and the designed classifiers.

in turn creates differences in data distributions. As mentioned before, the model trained on one domain cannot work on another one. A possible solution is to train models for each domain. Obviously, it is a big cost of human labor to label training data and domain-specific models are not appropriate for practical application. Considering that a well-trained classifier contains the defect detection knowledge on its training domains, we can transfer it to a new domain though no annotations are available for now.

Specific to the industrial scenarios, the two major categories of crystalline silicon cells include monocrystalline and polycrystalline cells, which have more obvious surface differences and belong to different domains. More impurities on the surface of polycrystalline cells make it harder to identify cracks, causing trouble in collecting polycrystalline labels. It is more practical to transfer the detecting experience of monocrystalline products to polycrystalline products.

Fig. 2 shows the overall process of our method. We embed the transfer learning mechanism in the defect detection network. Normalized EL images are directly used as the model input. During training, all images can be divided into two parts: the set of labeled monocrystalline images, and a set of unlabeled polycrystalline images. When the unlabeled images have a similar distribution as the labeled ones, theoretically, the model trained with labeled image set is also applicable to another domain that has the same data distribution. However, this is not always the case, so our approach is focusing on the scenario where the distributions of unlabeled and labeled images have gaps. This is part of the reason we choose to transfer the knowledge of labeled monocrystalline source domain to unlabeled polycrystalline target domain, the obvious surface differences increase the differences in data distributions. The model is trained with both labeled and unlabeled images jointly, using a simple CNN as the baseline. With the help of gradient reversal layer (GRL) (Ganin and Lempitsky, 2015) as the knowledge transfer method. In order to improve the efficiency of transfer learning, we add attention module to CNN and use it as the basis of knowledge migration. In addition, we add the class-specific domain classifier as a way to implicitly model the domain information of different classes, which is also beneficial to the knowledge transfer process.

After the training is completed, the overall goal is to allow the model to maintain the classification ability on the labeled monocrystalline image set while improving the defect detection ability on the unlabeled polycrystalline set. The overall approach is simple, effective and easy to follow.

Table 2
Used symbols.

Symbol	Definition
p_{pos}	Positive prediction of a sample
p_{neg}	Negative prediction of a sample
L_c	Defection classification loss function
L_{cls}	Defection classification loss
L_d	Domain classification loss function
L_{dom}	Domain classification loss
$L_{att-dom}$	Attention-based domain classification loss
L_{ca-dom}	Class-aware domain classification loss

3.1. Problem formulation

The problem is formulated based on the general setting of unsupervised domain adaptation. Specifically, given a set of labeled data $\mathcal{D}_s = \{\mathbf{x}_i, y_i\}_{i=1}^{n_s}$ for monocrystalline source domain and unlabeled data $\mathcal{D}_t = \{\mathbf{x}_i\}_{i=1}^{n_t}$ for polycrystalline target domain, we assume that they come from two marginal distributions $\mathcal{S}$ and $\mathcal{T}$. d denotes the domain label, i.e., $d_i = 1$ if $(\mathbf{x}_i, y_i) \sim \mathcal{S}$ and $d_i = 0$ otherwise. Our aim is to construct a classifier $\mathcal{F}(\mathbf{x}_i; \theta)$, which is able to complete classification task on data under the distribution of both $\mathcal{S}$ and $\mathcal{T}$. With the help of source supervision, the model can minimize the empirical loss for the source domain data, reducing the source risk $\epsilon_s(\mathcal{F}) = \Pr_{(\mathbf{x},y)\sim\mathcal{S}}(\mathcal{F}(\mathbf{x};\theta) \neq y)$. Combined with transfer learning, the classifier can learn transferable features as well, boosting the minimization of the target risk $\epsilon_t(\mathcal{F}) = \Pr_{(\mathbf{x},y)\sim\mathcal{T}}(\mathcal{F}(\mathbf{x};\theta) \neq y)$. Some other symbols used in this paper are illustrated in Table 2.

3.2. Classification using convolutional neural networks

CNN has been widely used as a powerful feature extractor in recent years. In our method, we utilize the ResNet-50 pre-trained on ImageNet to process the input image. Loading ImageNet pre-train parameters can be seen as a way of transfer learning. The empirical classification knowledge on ImageNet is exploited by parameters copying and fine-tuning. In the meanwhile, parameter loading is proved to be a better way of model initialization, which is used in other vision tasks, such as object detection and instance segmentation.

Since we are dealing with the classification task on a new domain, the last fully connected layer in the original ResNet-50 is replaced with a 2048-neuron in and 2-neuron out fully connected layer. It is usually called the classifier part relative to the whole network. The two output neurons are called *logits*, which measure the probability of having or

not having the micro-crack. Given an input image, we follow the same image processing procedure as ImageNet pre-train model to manipulate it. The reason is that parameter loading is only useful when the pre-trained model and the fine-tuning process adopt the same input data processing strategy. To be specific, the input image is first resized to 224×224, and normalized with the ImageNet dataset pixel means and standard deviations, which are calculated in the spatial dimension. The normalization method is to subtract the pixel means and divided by pixel standard deviations for each channel. Next, the processed image is sent to the classification network and output two logits: s_{pos} and s_{neg}. *pos* and *neg* denotes samples with micro-crack and undamaged samples, respectively. We use SoftMax to obtain the final probability, which is calculated by Eqs. (1) and (2).

$$p_{pos} = \frac{e^{s_{pos}}}{e^{s_{pos}} + e^{s_{neg}}} \tag{1}$$

$$p_{neg} = \frac{e^{s_{neg}}}{e^{s_{pos}} + e^{s_{neg}}} \tag{2}$$

The classification model is end-to-end trained by minimizing standard multi-class cross entropy loss, as shown in Eq. (3).

$$L_{cls} = -y_{pos} \log p_{pos} - y_{neg} \log p_{neg} \tag{3}$$

y_{pos} and y_{neg} in Eq. (3) are training labels. For a positive sample, we have $(y_{pos}, y_{neg}) = (1, 0)$ and for a negative sample, $(y_{pos}, y_{neg}) = (0, 1)$.

As the micro-crack detection task can be formulated as a binary classification problem, it seems more natural to use a single-neuron out fully connected layer to get probability predictions and utilize binary cross entropy as the loss function. But we adopt the idea of multi-class classification is to deal with this problem, although they are mathematically equivalent. The reason is that we want to separate the positive and negative samples in the feature space. In this way, we can perform class-disentangled knowledge transfer for different types of samples, which will be introduced in 3.3.3.

3.3. Unlabeled domain classification with transfer learning

As mentioned before, for labeled monocrystalline source domain data, CNN can be used to complete the training of a classification model. Even if data from the same distribution is collected in some way later, we can use the existing classification model to provide pseudo-labels and add them to the training sample. However, when the newly collected polycrystalline data comes from a new production line based on other different cell technology, the previous classifier is completely invalid on this batch of data, so it cannot be trained directly utilizing pseudo-labeling. The idea of solving this problem is to use transfer learning and jointly train a classifier using the labeled data from the monocrystalline source domain together with the unlabeled data from the polycrystalline target domain. This can be seen as a way to align the features between the source and target domains, and thus migrate the classification ability on the source domain to the target domain.

In this section, two general transfer methods will be introduced first. Then, we propose two new components: attention-based transfer learning and class-specific domain classifier, to further boost the performance of knowledge transfer for micro-crack detection on EL images of PV modules.

3.3.1. Defect knowledge transfer

Most transfer learning methods are focusing on the learning of transferable features. This idea is quite natural. Since the classifier fails due to the inconsistent distribution of the data in the monocrystalline source and polycrystalline target domains, it is a good choice to align the feature distributions of the two domains during the feature extraction process. Gradient Reverse Layer (GRL) is a typical approach, which forces the parameters to update in the direction of confusing the features of both domains by reversing the gradient.

To manipulate the gradient in the right way, a domain discriminator is added to the same position as the classifier part. The introduction of the domain discriminator is for distinguishing between the source domain and the target domain. Now our whole model is composed of three parts: the feature extractor $\mathcal{F}_f(\mathbf{x}; \theta_f)$, the classifier $\mathcal{F}_c(\mathbf{x}; \theta_c)$ and the domain discriminator $\mathcal{F}_d(\mathbf{x}; \theta_d)$, where θ_f, θ_c and θ_d are corresponding model parameters. For classification task, we can train the network by minimizing the classification loss $L_c(\mathcal{F}_c(\mathcal{F}_f(\mathbf{x}_i; \theta_f); \theta_c), y)$, where y is the category label and L_c refers to the cross entropy loss function. For domain distinguishing task, if we want the domain discriminator to be functional, it should be trained by minimizing the discriminator loss $L_d(\mathcal{F}_d(\mathcal{F}_f(\mathbf{x}_i; \theta_f); \theta_d), d)$, where d is the domain label and L_d is the binary cross entropy loss.

However, feature would be very discriminative in this case, which is exactly the opposite of the purpose that we want to narrow the distribution divergence of features between different domains. Considering that the domain discriminator has been well trained and its parameters fixed, in order to align the domain feature distribution, we can confuse the domain discriminator by maximizing the discriminator loss. From the perspective of gradient, it is equivalent to reversing the gradient and minimizing the domain loss.

Following this idea, the whole model can be end-to-end trained with stochastic gradient decent (SGD) updates. Assuming that a sample $\mathbf{x}_i$ is input to the model for the forward pass and classification loss L^i_{cls} and domain loss L^i_d is computed, parameters of each part are updated as Eqs. (4), (5) and (6).

$$\theta_c \leftarrow \theta_c - \mu \frac{\partial L^i_{cls}}{\partial \theta_c} \tag{4}$$

$$\theta_d \leftarrow \theta_d - \mu \frac{\partial L^i_d}{\partial \theta_d} \tag{5}$$

$$\theta_f \leftarrow \theta_f - \mu \left(\frac{\partial L^i_{cls}}{\partial \theta_f} - \lambda \frac{\partial L^i_d}{\partial \theta_f} \right) \tag{6}$$

In Eqs. (4), (5) and (6), μ denotes the learning rate and λ denotes the gradient reverse weight.

More formally, the operation of gradient reverse can be defined as a function $R_\lambda(\mathbf{x})$. The forward and backward behavior are described in Eqs. (7) and (8).

$$R_\lambda(\mathbf{x}) = \mathbf{x} \tag{7}$$

$$\frac{\partial R_\lambda}{\partial \mathbf{x}} = -\lambda \mathbf{I} \tag{8}$$

In Eq. (8), $\mathbf{I}$ is the identity matrix. With this self-defined function, the total loss function is computed as Eq. (9).

$$L = L_c(\mathcal{F}_c(\mathcal{F}_f(\mathbf{x}_i; \theta_f); \theta_c), y) + L_d(\mathcal{F}_d(R_\lambda(\mathcal{F}_f(\mathbf{x}_i; \theta_f)); \theta_d), d) \tag{9}$$

In the inference stage, the domain discriminator can be removed from the model. Only the feature extractor and classifier are needed for defect detection.

3.3.2. Attention-based transfer learning

Attention mechanism is a popular approach that can help the network focus on meaningful areas. From another point of view, attention can be seen as a form of encoding effective information in the network. For the micro-crack detection network in our method, we propose to transfer the knowledge in channel attention in addition to the migration in the final feature. This approach can align the distribution of channel attention in feature maps and further reduce the feature distribution gap between different domains.

First, we apply the attention mechanism (Hu et al., 2018) to the last three blocks of ResNet-50. The detail of changing one block is illustrated in Fig. 3. For a standard residual block output before shortcut

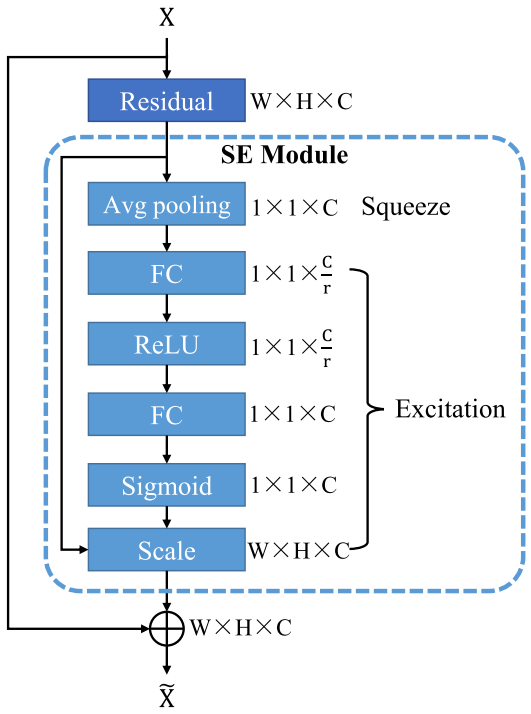


Fig. 3. The structure of the attention mechanism module (SE module) and its combination with a block in ResNet-50. The attention mechanism module (SE module) is embedded into the residual learning branch in the residual structure, and its dimension reduction coefficient r is 16. The module combines the residual feature with the shortcut after performing squeeze, excitation and scale operations.

$\mathbf{U} \in \mathbb{R}^{H\times W\times C}$, it is first squeezed spatially. If we write the output as $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_C]$, the squeezed vector $\mathbf{z} \in \mathbb{R}^C$ can be calculated by Eq. (10).

$$z_c = \sum_{i=1}^{H}\sum_{j=1}^{W} u_c(i,j) \tag{10}$$

Next, a process of adaptive recalibration is performed on $\mathbf{z}$. Its dimension is reduced by r times with a non-linear transformation, which is shown in Eq. (11).

$$\mathbf{o} = \delta(\mathbf{W}_1\mathbf{z}) \tag{11}$$

In Eq. (11), $o \in \mathbb{R}^{\frac{C}{r}}$ denotes the output, $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r}\times C}$ is the learnable transformation matrix and δ refers to the ReLU function. Then the dimension of the attention vector is recovered to C by a linear transformation, and a simple gating mechanism is performed with a sigmoid function. The calculation can be formulated as Eq. (12).

$$\mathbf{s} = \sigma(\mathbf{W}_2\mathbf{o}) \tag{12}$$

In Eq. (12), $s \in \mathbb{R}^C$ denotes the output, $\mathbf{W}_2 \in \mathbb{R}^{C\times\frac{C}{r}}$ is another learnable transformation matrix and σ refers to the sigmoid function. Finally, the recovered channel attention $\mathbf{s}$ is used to scale the residual output $\mathbf{U}$. The output equipped with attention is computed as Eq. (13).

$$\hat{\mathbf{u}}_c = s_c\mathbf{u}_c \tag{13}$$

After applying the attention mechanism, another indicator $\mathbf{o}$ can be used for attention knowledge transfer. As shown in Fig. 2, we add domain discriminators to attention blocks. For the ith attention intermediate result $\mathbf{o}_i$, we use $\mathcal{F}^i_{att-dom}$ as its new domain discriminator. The domain loss can be computed as Eq. (14).

$$L_{att-dom} = L_d(\mathcal{F}^i_{att-d}(R_\lambda(\mathbf{o}_i); \theta^i_{att-d}), d) \tag{14}$$

The training strategy totally follows the gradient reverse strategy mentioned before. The domain discriminator introduced in 3.3.1 is a useful tool to align features of different domains. However, it is class-agnostic. In the data batch for a single training iteration, the proportion of positive and negative samples is uniform, no matter for the source domain or the target one. It may cause unreasonable sample feature transfer. For example, features of positive samples from the monocrystalline source domain are forced to be aligned with the features of negative samples from the polycrystalline target domain. To address this issue, we add class-aware domain classifiers to our model.

Specifically, two extra domain discriminators are inserted into the original position, which is shown in Fig. 4.

3.3.3. Class-aware domain discriminator

They make domain predictions for positive and negative samples in a soft way. Since class labels are not available for EL images in polycrystalline target domains, we use predictions of the classifier as a class indicator. To unify the training of two domains, class scores are adopted to weight the domain loss. We use $\mathcal{F}_{pos-d}(\mathbf{x}; \theta_{pos-d})$ and $\mathcal{F}_{neg-d}(\mathbf{x}; \theta_{neg-d})$ to denote the domain classifiers for positive and negative samples, respectively. The new domain loss can be calculated as Eq. (15).

$$\begin{aligned} L_{ca-dom} = {} & p_{pos} \cdot L_d(\mathcal{F}_{pos-d}(R_\lambda(\mathcal{F}_f(\mathbf{x}_i; \theta_f)); \theta_{pos-d}), d) \\ & + p_{neg} \cdot L_d(\mathcal{F}_{neg-d}(R_\lambda(\mathcal{F}_f(\mathbf{x}_i; \theta_f)); \theta_{neg-d}), d) \end{aligned} \tag{15}$$

During training, p_{pos} and p_{neg} are also contained in classification loss. If we do not do anything to them, the class-aware domain loss would push them to 0 in optimization, which may have a conflict with the classifier loss. So the gradient of p_{pos} and p_{neg} in Eq. (15) needs to be cut in the backward propagation. In this way, they can be considered as loss weights and do not have a direct effect on classifier loss optimization. The overall loss is computed as Eq. (16).

$$L = L_{cls} + \lambda_1 L_{att-dom} + \lambda_2 L_{ca-dom} \tag{16}$$

In Eq. (16), λ_1 and λ_2 are loss weights.

It should be noted that at the start of training, the class prediction is biased. So it is unreasonable to use p_{pos} and p_{neg} as the guidance for the use of domain discriminators. To solve this problem, we choose to use a class-agnostic domain discriminator at first. When the class prediction becomes more accurate, class-aware domain discriminators are introduced and initialized with the class-agnostic weights. With this trick, it is avoided that class predictions are collapsed to one specific category.

4. Experiments

4.1. Dataset and evaluation metrics

In order to meet the problem setting of UDA, we use the academic solar cell dataset proposed by Deitsch et al. (2019) and the dataset provided by an industrial production line to simulate products manufactured according to different technical solutions. We construct labeled source domain data and unlabeled target domain data as shown on the left and right sides of Fig. 1, respectively, and try to apply the model trained based on monocrystalline data to polycrystalline data. The public dataset proposed by Deitsch et al. (2019) is composed of 2624 aligned solar cell images extracted from high-resolution EL images, which can be divided into two categories according to monocrystalline silicon solar cells and polycrystalline ones, and each category also contains many types of cell products. This dataset uses floating-point numbers to represent the possibility of micro-cracks in each sample. We extract all monocrystalline samples with the possibility determined as 0.0 or 1.0 in the dataset for experiments. A total of 901 images, including 313 cracked images and 588 images without cracks. We divide 901 monocrystalline images by 8:2 to form the training set and verification set in the model training stage, respectively, that is, the training set contains 720 images and the verification set contains 181

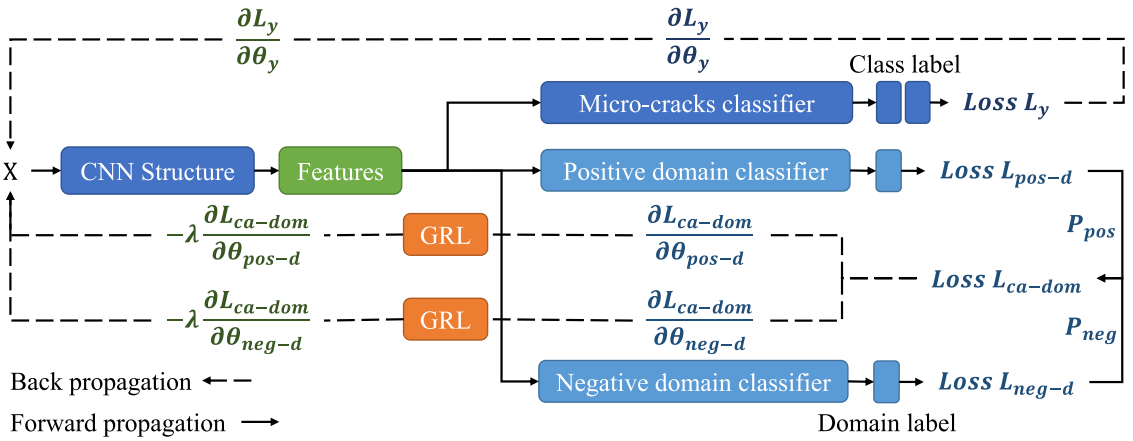


Fig. 4. The insertion position, forward propagation and backpropagation process of class-aware domain classifiers. GRL is used to realize gradient inversion, optimize the objective function in the negative gradient direction, and promote the alignment of the source domain and target domain features of positive and negative classes, respectively.

images. The unlabeled target domain dataset we used in the experiment consists of 794 EL images of polycrystalline cells, including 281 and 513 images with and without micro-cracks, respectively. As shown in Fig. 1, the shapes and positions of micro-cracks with shallow traces are diverse. There are different numbers of main grids on the surface of these two solar cell products, the polycrystalline products not only have different data distribution from monocrystalline products, but also many impurities on their surfaces will interfere with micro-crack detection. It is a practical task in many transfer directions of similar products and different products.

In real industrial production scenarios, the cracked solar cells are difficult to collect, so the number of samples is relatively small. Therefore, there exists sample imbalance in both the source domain dataset and target domain dataset. In order to more truly reflect the model's ability to detect cracks, attention should be paid to avoid the accuracy being compromised by the majority of samples while masking the problem of weak detection ability for cracked data. We choose to introduce recall and precision as evaluation metrics to examine the recognition ability and accuracy of the model for cracked samples. And we adopt the harmonic mean F1-score of recall and precision as a comprehensive measure of the ability of the model.

4.2. Implementation details

Our experiments are based on the open-source framework PyTorch and a GeForce GTX 1080Ti graphics card. We adopt ResNet-50 as the backbone network for the experimental model, uniformly adjust the dimensions of the EL images to 224 × 224 × 3, and input the images into the convolutional network after normalizing the three channels. In the training phase, the batch size is set to 32, and to fully learn information from limited samples, we introduce simple data augmentation methods to flip input EL images horizontally or vertically with a probability of 0.5. We update the parameters of a convolutional neural network (feature extraction network) with the SGD algorithm with an initial learning rate of 0.01, momentum of 0.9, and weight decay of 0.001. The training stage contains a total of 80 epochs, the learning rate is adjusted in epoch to be gradually reduced. We take the parameters of the model when the training loss gradually converges to stabilize as the final training result. All our deep learning models are trained based on the experimental configurations described above.

4.3. Ablations

4.3.1. Impact of adversarial learning

Due to the neat surface of the monocrystalline cell and the high impurity of the polycrystalline cell, if the model trained on the source domain is applied directly to the target domain, most of the impurities on the cell surface will be wrongly recognized as cracks. It will cause the problem of recalling the cracked cells and the normal cells at the same time. The recall is falsely high and the precision is very low. This also indicates that the model does not have the ability to identify micro-cracks in the target domain, and the basis for judging the existence of cracks is not suitable. Therefore, only considering the recall may mask the problem that the model misjudges numerous normal cells as cracked. So we choose F1-score to measure the recall and precision at the same time.

Table 3
Ablation results with attention-based discriminators (AD) and class-aware discriminators (CD).

Methods	Recall	Precision	F1-score
GRL	84.34%	63.54%	0.7248
GRL+AD	80.78%	**91.16%**	0.8566
GRL+CD	77.22%	75.35%	0.7627
GRL+AD+CD	**84.70%**	90.15%	**0.8734**

As shown in the first two rows in Table 4, both the model obtained by the traditional ML method Support Vector Machines (SVM) and the deep learning model with ResNet50 as the backbone perform worse. After training on the source domain, the model with ResNet-50 as the backbone can achieve an F1-score of 0.7143. However, the F1-score of this model directly applied to the target domain is only 0.6103. This CNN model can be regarded as the baseline. We align the feature distributions of the historical labeled source and unlabeled target domain using GRL based on adversarial learning and a simple-structured domain classifier. We input the final feature of ResNet-50 into the domain discriminator to distinguish source and target domains, after gradient reversal of GRL to confound the feature distributions of the two domains, mitigating the effect of inconsistent data distributions. As shown in the third row of Table 4, by aligning the final features of the two domains, the adaptability to the target domain of the model is significantly enhanced. The F1-score has increased by 0.1145 and the precision has increased by 18.59%, indicating that the misjudgment of the model on the normal solar cell is reduced.

The above experiments show that the GRL method using the idea of adversarial learning can narrow the feature distribution of the two domains, which can promote the model to pay attention to the information that is helpful to identify micro-cracks across domains.

4.3.2. Impact of attention transfer

We add the attention mechanism structures (SE modules) to the three blocks of the last stage of ResNet-50. The attention mechanism focuses on the channel dimension and adjusts the weight given to each

Table 4
Comparison of model detection abilities between SVM, CNN, and final feature migration.

Methods	Recall	Precision	F1-score
SVM	79.00%	48.79%	0.6033
CNN	**95.02%**	44.95%	0.6103
CNN+GRL	84.34%	63.54%	0.7248
Our method	84.70%	**90.15%**	**0.8734**

Table 5
Ablation results of migration locations of attention features after the introduction of attention mechanism. '4_1', '4_2', '4_3' correspond to the attention features of the first, second and third block of the fourth stage of ResNet-50, respectively.

Transfer position			Recall	Precision	F1-score
4_1	4_2	4_3			
			90.04%	62.16%	0.7355
✓			91.19%	64.97%	0.7585
	✓		86.48%	69.83%	0.7727
		✓	85.77%	69.45%	0.7675
✓	✓		85.05%	79.14%	0.8199
	✓	✓	82.56%	87.88%	0.8514
✓		✓	74.02%	**97.65%**	0.8421
✓	✓	✓	80.78%	91.16%	**0.8566**

Table 6
Influence of migrating attention features on the detection ability of models that introduce attention modules. 'GAP' indicates global average pooling.

Transfer content	Recall	Precision	F1-score
CNN w. Max pooling	**93.59%**	60.32%	0.7336
CNN w. GAP	82.21%	81.34%	0.8177
Attention	80.78%	**91.16%**	**0.8566**

channel feature according to the input, to make the network focus on the features that help to distinguish micro-cracks and suppress useless features. After introducing the attention mechanism, the experiment is carried out based on the network structure ResNet-50 + SE modules. As shown in the first row of Table 5, compared with CNN + GRL, the F1-score increased by 0.0107 after introducing three SE modules into the backbone network.

The introduction of the attention mechanism provides more choices for feature transfer. We conducted ablation experiments on the feature output of the attention mechanism as shown in Table 5 to gradually enhance the intensity of feature transfer based on the final feature transfer. From the overall results, the deeper the location of transfer features in the network and the richer the transfer features, the better the performance of the model in the target domain. For example, row 4 in Table 5 represents the feature of further migrating the output of the last attention mechanism module (SE module), which can improve the F1-score by 0.032 compared with the feature of not migrating attention features. The F1-score of 0.8566 can be achieved by migrating the three SE modules at the same time. Compared with not migrating the attention mechanism features, the F1-score is increased by 0.1211 from the perspective of aligning the distribution of channel attention in feature maps.

In order to further verify the influence of adding other features of transfer on the detection ability of the model and the advantages of the attention-based transfer, we conducted ablation experiments as shown in Table 6. In the network structure with attention mechanism, we consider migrating the features of the residual part of the last three blocks of ResNet-50. To meet the input of GRL, we try to convert the feature map into feature vector by max pooling and global average pooling (GAP). Adding features at the same location for transfer, the result of GAP transformation is more suitable for the model of this task, and F1-score is 0.0841 higher than Max pooling. The feature vector transformed by GAP can retain more cell surface information than that transformed by Max pooling, which is helpful to extract the micro-crack feature.

Enriching the features of migration helps the model adapt to the target domain, but attention features transfer still maintains a leading advantage compared with these two methods, and the F1-score increases by at least 0.0355. Thus, we choose to further migrate the three attention mechanism features based on the transfer of final features, to make the attention mechanism of the model focus on the features suitable for both domains at the same time. As shown in Table 3, the GRL + AD method can increase F1-score to 0.8566, an increase of 0.1318 compared with GRL.

4.3.3. Impact of class-aware domain discriminator

In the above experiments, each GRL corresponds to one class-agnostic domain classifier, which ultimately obfuscates the two domains and reduces the distribution differences of features between different domains of the model output. However, aligning features across the entire domain will result in a forced alignment between the two kinds of products that need to be categorized as cracked or not, which is not conducive to crack detection. Therefore, we consider aligning the features of the same class of cells between different domains. The third row of results from Table 3 corresponds to upgrading the original single class-agnostic discriminator of the GRL into two class-aware discriminators (CD) under the ResNet50 structure. Under this network structure, the CD can elevate the F1-score by 0.0379. As shown in Table 7, compared with a single class-agnostic discriminator, two class-aware discriminators can increase the recall by 3.92% while maintaining a similar precision level. It can be seen that the domain discriminators responsible for the positive and negative classes can align the features between domains without disturbing the classification.

We use the predictive logits after Softmax to guide the training of two class-aware discriminators, so the sum of the weights assigned to the two discriminators is 1. To verify that the improvement of model detection ability comes from the class-aware idea rather than simply adding discriminators, we tested the effect of simply applying two class-agnostic discriminators to the same location. During training, the weight assigned to the two discriminators is 0.5, and the sum is also 1. As shown in the second row of Table 6, under this discriminator combination scheme, the F1-score of the model is 0.8211, which is 0.0355 lower than that of one single class-aware discriminator. When two class-aware discriminators are selected, the weights of different values guide the discriminator to focus on the domain discrimination of the corresponding class to ensure feature alignment in the same class. However, by simply adding the class-agnostic discriminator, the smaller weight has only a weak updating effect on the parameters of the two discriminators. Compared with a single classifier, the training is insufficient, resulting in a decline in the results. Therefore, it is not significant to simply increase the number of discriminators, but the introduction of the class-aware idea can improve the model detection ability.

The above experimental results are carried out under the configuration that the weight of GRL output loss corresponding to each part of the features is equal. As shown in Table 3, GRL + AD+CD is carried out under the configuration of $\lambda_1 = 0.25$ and $\lambda_2 = 0.25$. We tested the model effect under other weights combinations, as shown in Tables 8 and 9. Fix parameters λ_2 and λ_1 respectively and adjust another parameter in steps of 0.05.

We still take the experimental results when both weights λ_1 and λ_2 are 0.25 to represent the general level. In Table 3, the F1-score of GRL + AD + CD increased by 0.2631 relative to CNN baseline and 0.1486 relative to GRL.

4.4. Qualitative results

4.4.1. Visualization of activation maps

To intuitively observe the detection effect of the model, we visualize the last convolutional layer of the model's network structure. Fig. 5 allows us to observe the response of the corresponding feature mapping

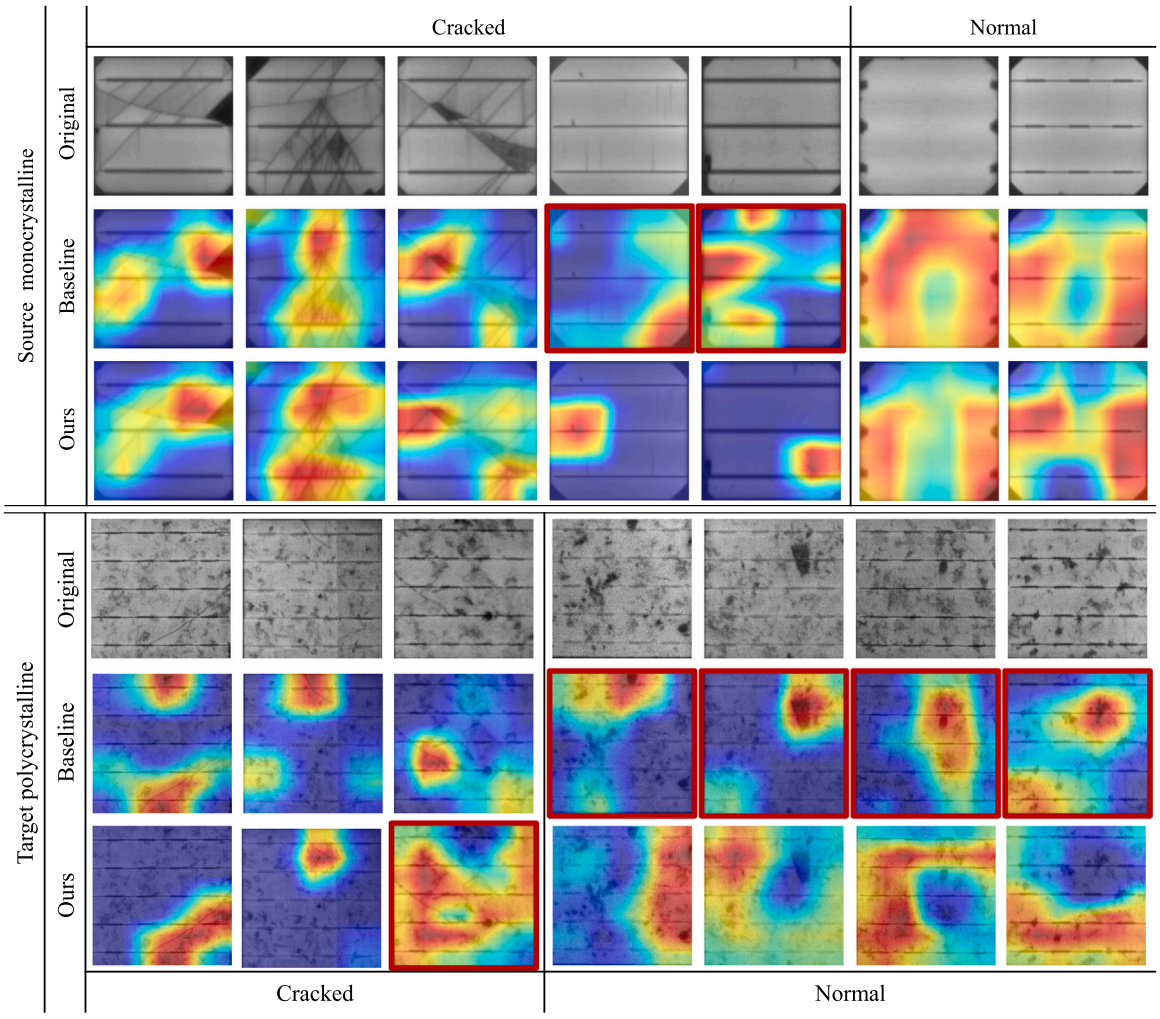


Fig. 5. Visualization of the last convolutional layer feature maps of the models. The top and bottom sections of the diagram show the qualitative results of the model on the source domain monocrystalline validation set and target domain polycrystalline test set. Each section contains three rows, the first is the original picture of the cell, and the next two rows correspond to the baseline and our model, respectively. The horizontal direction of this diagram indicates whether the cell is cracked. Feature maps marked by a red border belong to false predictions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7
Influence of domain classifier design on model detection results.

Class-aware	N_{dom}	Recall	Precision	F1-score
×	1	80.78%	**91.16%**	0.8566
×	2	77.58%	87.20%	0.8211
✓	2	**84.70%**	90.15%	**0.8734**

Table 8
Change of model detection effect with loss weight λ_1 when fixed weight λ_2 is 0.25.

λ_1	Recall	Precision	F1-score
0.10	88.26%	86.71%	0.8748
0.15	83.27%	94.35%	0.8847
0.20	77.94%	96.48%	0.8622
0.25	84.70%	90.15%	0.8734
0.30	79.36%	94.49%	0.8627
0.35	82.92%	93.95%	0.8809
0.40	79.22%	97.39%	0.8767

for the two types of cells with or without cracks, and to compare the differences in response levels at different locations on the feature maps of baseline and our model. The detection capabilities of these two models correspond to the second and last row of the Table 4, respectively.

Table 9
Change of model detection effect with loss weight λ_2 when fixed weight λ_1 is 0.25.

λ_2	Recall	Precision	F1-score
0.10	80.78%	91.53%	0.8582
0.15	79.72%	95.73%	0.8699
0.20	80.78%	94.58%	0.8714
0.25	84.70%	90.15%	0.8734
0.30	80.78%	93.42%	0.8664
0.35	79.72%	94.92%	0.8665
0.40	81.49%	100.0%	0.8980

In the visualization results of Fig. 5, the position where the color is biased towards red plays a greater role in obtaining the prediction results. It can be seen from the overall observation that as long as there is a micro-crack, it can be judged that the cell is cracked while judging that there are no cracks, we need to pay attention to the state of the whole cell surface. After training, both models can achieve better results on the source domain validation set, but the more capable model can locate crack locations more accurately and without missing. There is a big gap between the performance of these two models on the target domain test set. The polycrystalline impurities in the test set will interfere with defect detection. Our model can learn the basis for correctly distinguishing cracks and agree that there are impurities on

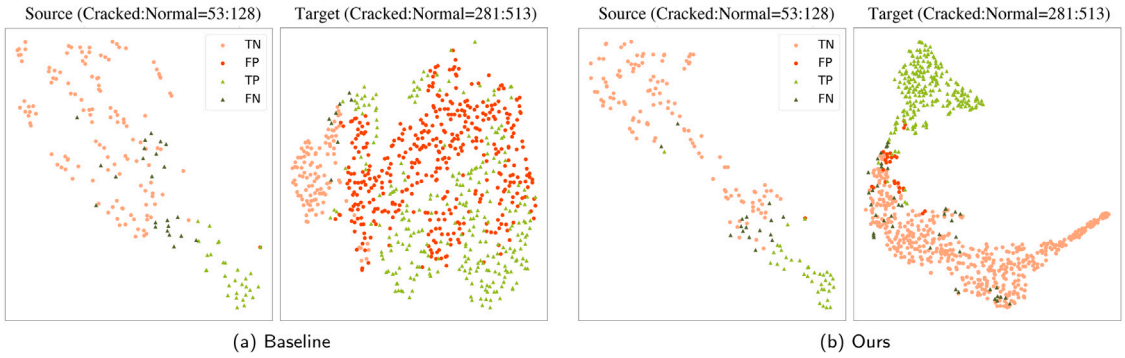


Fig. 6. Visualization of the features used for classification of the baseline and our model after t-SNE dimensionality reduction. The two graphs in each group correspond to the source domain verification set and target domain test set, respectively. TN, FP, TP and FN in the graph represent true negative class, false positive class, true positive class and false negative class, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 10
Comparison of our method with some popular migration methods. 'Forward' indicates the input required for the method to calculate the loss, where 'label' indicates the real label and 'logits' indicates the output predicted by the model.

Methods	Forward		Recall	Precision	F1-score	Reference
	Source	Target				
DANN	Features	Features	84.34%	63.54%	0.7248	Ganin and Lempitsky (2015)
MMD	Features	Features	80.07%	80.36%	0.8021	Long et al. (2015)
BNM	–	Softmax logits	**89.32%**	76.76%	0.8257	Cui et al. (2020a)
DSAN	Features(label)	Features(logits)	70.82%	54.67%	0.6171	Zhu et al. (2020)
DeepCoral	Features	Features	83.27%	53.06%	0.6482	Sun and Saenko (2016)
DAAN	Features(logits)	Features(logits)	84.34%	52.55%	0.6475	Yu et al. (2019)
Ours	Features(label)	Features(Stable logits)	84.70%	**90.15%**	**0.8734**	–

the surface of normal cells. While the baseline without migration will judge most of the impurities as cracks, resulting in the misjudgment of normal cells as defective cells, which is the phenomenon that the model fails to adapt to the data distribution of a batch of new products.

4.4.2. Analysis of feature space

We use t-distributed stochastic neighbor embedding (t-SNE) to carry out nonlinear dimensionality reduction on the features to be input into the crack classifier and visualize the distribution of dimensionality reduction features in the two-dimensional coordinate system. In Fig. 6, green corresponds to cracked cells, red corresponds to normal cells, and dark red and dark green are used to mark the samples with the wrong prediction. The features' distribution of the baseline and our model in the source domain is similar. Both models can learn similar information from the training set to identify whether there are cracks. However, our model can bring the feature distribution of the same class closer, and the clustering effect is more obvious. As shown in Fig. 6(a), on the target domain, there are many dark red dots in the visualization of the baseline, indicating that most of the normal cells have been misjudged. The class with cracks occupies most of the feature space, resulting in virtual high recall and low precision. This indicates that the baseline model fails to correctly segment the feature space and is not suitable for new batches of products. As shown in Fig. 6(b), our model can adapt to the products of the source domain and target domain at the same time by using AD and CD, learning the features suitable for identifying cracks in the target domain.

4.5. Comparisons

Table 10 shows the performance results of our method and some popular transfer methods in the target domain. Among the comparison methods, the BNM with better effect does not require source domain data and restricts the prediction logits of the model to the target domain. The F1-score of this method can reach 0.8257, and the F1-score of our method is 0.0477 higher than that of this method.

Methods with an F1-score below 0.7 include DSAN, DAAN and DeepCoral. DeepCoral uses the Frobenius norm between the source domain and target domain to measure the distance between these two domains. As the same method to calculate and narrow the features' distribution between domains, MMD uses multiple Gaussian kernels to calculate higher moments to describe the distribution of domains, and then calculate the distance between domains. The F1-score of this method can reach 0.8021. It can be seen that DeepCoral is not as suitable for the crack detection task in this paper as MMD. The other two methods, DSAN and DAAN, also have their strengths, and we consider that their failure to perform better is due to the additional reference to the predicted logits of the target domains when calculating losses. The micro-crack detection task in this paper simultaneously has the characteristics of different data distributions, unrecognizable cracks and impurity interference. At the initial stage of features transfer, these reasons act in parallel to cause the model to misjudge impurities as cracks, the model's ability to identify cracks is greatly reduced, and the predicted logits are inaccurate and unstable. These reasons are likely to make it difficult for the two methods to get a good initial state or even deadlock. We consider introducing the predicted logits into the loss measurement phase after they are stable to alleviate this problem, increasing F1-score by 0.2563 and 0.2259, respectively, compared with these two methods.

4.6. Discussion

The above ablation experiments can show that our method is reasonable and effective. As shown in Table 3, the introduction of GRL can effectively alleviate the problem that the model loses the classification ability in the new target domain and misjudges numerous normal cells as defective. And increase F1-score to 0.7248, an increase of 0.1235 compared with CNN.

The introduction of the attention mechanism provides more options for feature alignment. Migrating deeper attention features or adding

attention features used for transfer both facilitate feature alignment. We can still explain from Table 6 that the improvement of model ability mainly comes from the transfer of the features of the attention mechanism. We finally chose to simultaneously migrate the three part attention features and the final feature. Compared with GRL, F1-score can be improved by 0.1318.

As shown in the ablation experiment in Table 7, simply increasing the number of class-agnostic discriminators cannot improve the performance of the model, but considering the class-aware idea to align the features between domains based on positive and negative classes is more conducive to improving the effectiveness of the model. After introducing class-aware discriminators, the F1-score of the model can be further improved from 0.8566 to 0.8734.

The visualization results can also echo the above conclusions. Our method can locate the cracks more accurately and adapt the monocrystalline defect detection model to polycrystalline surfaces with more impurities. It can strengthen the clustering of source domain features, learn the judgment basis of micro-cracks on the polycrystalline surface, and reasonably divide the feature space of new products.

5. Conclusion

This work proposes a novel defect detection method for solar cells in EL images under the scenario of unsupervised domain adaptation, which specifically focuses on the problem that more impurities on the surface of polycrystalline cells lead to higher difficulty in crack detection and data labeling compared with monocrystalline cells.

A network with CNN architecture is trained on monocrystalline labeled data from the source domain and transfers the defect detection knowledge to the polycrystalline target domain with the help of unlabeled data from it. Equipped with adversarial learning, attention-based transfer learning and class-aware domain discriminator are further introduced to improve the transfer effect on EL solar cell images. Finally, the F1-score of the model in the polycrystalline target domain can be improved by 0.2631, reaching a recall of 84.70% and a precision of 90.15%, which can better adapt to a batch of new products.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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